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This procedure applies to 1.5Tand 1.0T, Signa systems.

Description - This material covers the MDS/PAC/SRI board Level tests, test LEDs that are available, error messages, and test descriptions.

Note

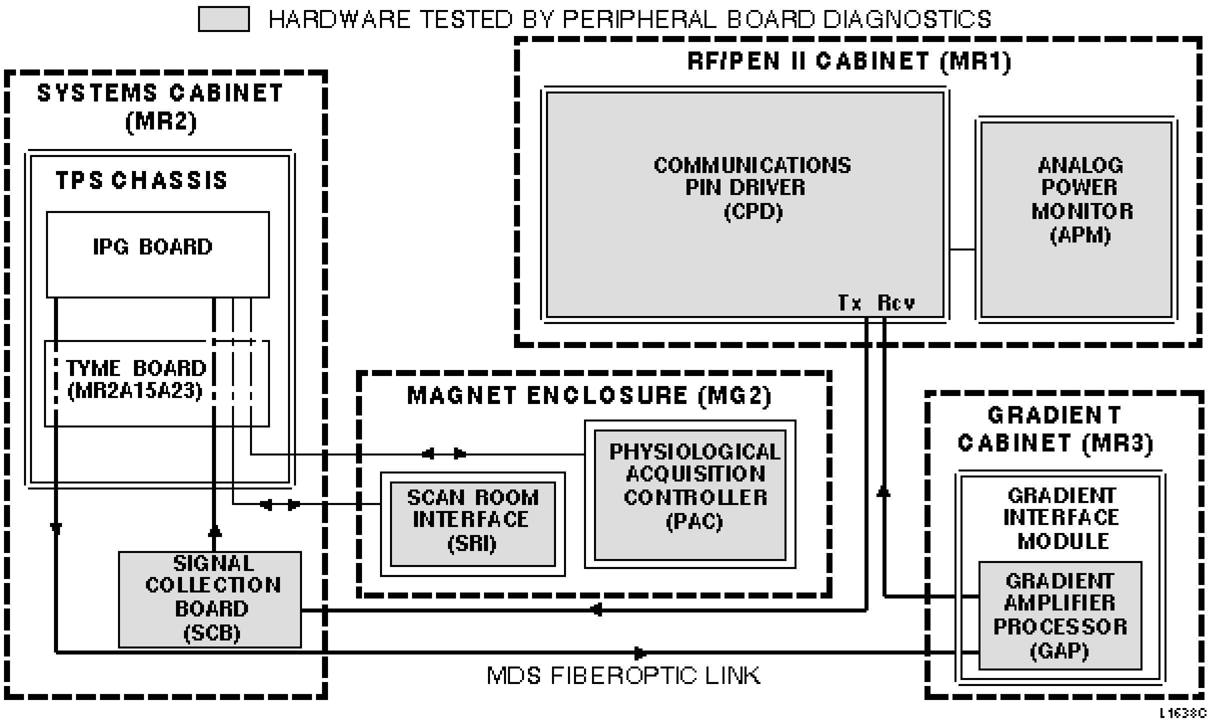
Do not start multiple copies of diagnostics. Multiple copies of diagnostics has caused different anomalies such as locked up diagnostic windows.

1- HARDWARE TESTS

MDS/PAC/SRI Board Level diagnostics are disk-based tests. Table 1-1 shows a list of tested boards; for the location of tested boards, see Illustration 1-1.

TABLE 1-1
TPS PERIPHERAL HARDWARE TESTED

Hardware	Test Menu
Gradient Driver Subsystem	MDS
Multicoil Driver Board	MDS
Physiological Acquisition Controller (PAC)	PAC
Power Monitor	MDS
RF Amplifier Controller Board	MDS
Scan Room Interface (SRI)	SRI
Serial Peripheral Converter (SPC)	MDS
Signal Collector Board (SCB)	MDS
TR/Dynamic Disable Board	MDS



TPS PERIPHERAL HARDWARE TESTED
ILLUSTRATION 1-1

2- STARTING BOARD LEVEL TESTS

The board level tests are all disk-based and can be selected/run only when Signa Horizon 8.X software is running. These tests are downloaded/run via the VME I/F board located in the TPS/ISE chassis. Status/error messages are returned to the host computer for logging/display as applicable.

Board level tests may be initiated individually, or in sets. Each procedure is described below.

Note

The gradient board level tests are available ONLY if the proprietary software was previously loaded from the Service or Customer Diagnostic Tape. A valid security key is also required to gain access to the complete set of tests.

1. When Signa software is up, from the Service Desktop, select **[Diagnostics]**, and **[Start]**. Wait for the Diagnostic Main Menu to appear.

2. Select **[Reset Test Selections]** to turn off all test selections.

3. Select iteration count: [1] (default), [iterations] (the number of which you may choose),or [Duration] (time that you may select).

4. Decide whether to run all default Peripheral Board Level tests, or just selected ones.

   To run all default Peripheral Board Level Tests:

   a. Select **[Full]** from the Test Selection Menu.
   b. Select **[Run Diags]**.

To run selected Peripheral Board Level Tests:

a. Select the desired Board Level Test icon (TPS/CPU, Host CPU, etc.) from the Diagnostics Menu. Wait for the submenu to appear.
b. Select the desired set of tests, (by clicking on the box next to the appropriate test).
c. Select **[Run Diags]**.

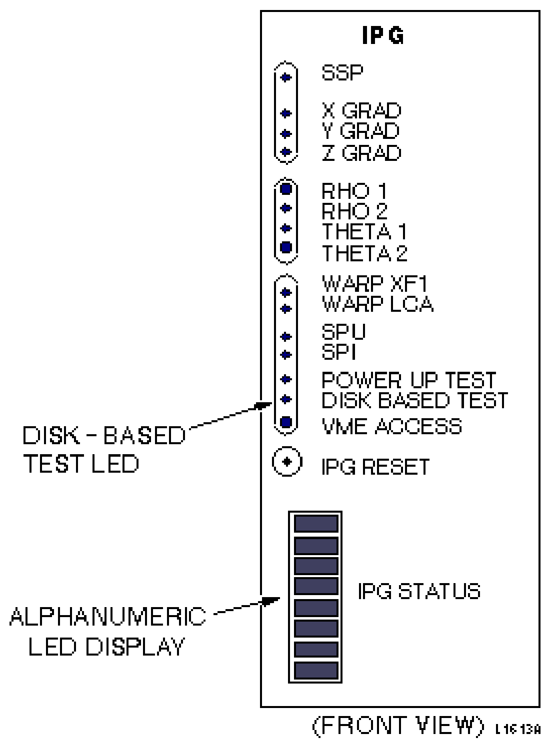
5. Select **[IPG]** and then **[MDS]** from the Submenu Selection. Wait for the MDS Menu to appear. Select the desired tests: (See section 4-3.)

6. Finally, select **[Run Diags]**.

3- TPS/ISE TEST INDICATORS

3-1 Local Test Indicators

Local test indicators are provided on the Integrated Pulse Generator (IPG) board of the TPS/ISE. One green LED (DISK BASED TEST) and an alphanumeric display (IPG STATUS) provide the status of TPS/ISE disk-based tests and general subsystem activity. These LEDs (see Illustration 3-1) provide backup error/status indication for gradient board level tests in case communications with the host computer fails. While the gradient board level tests are executing, the DISK BASED TEST LED lights when the IPG is accessed. The IPG STATUS display also shows the abbreviated name of the test in progress.



LOCATION OF IPG TEST INDICATORS
ILLUSTRATION 3-1

3-2 Operator Work Station Test Status Display

Diagnostic test status is also indicated in the Results window. At the bottom of the Results window, messages such as `Resetting TPS/IPG`, `Downloading <test name>`, `<test name> Running`, and `Diagnostics Completed` help indicate diagnostic test status.

3-3 Message Log

If an error occurs during board level tests, an error message is reported in the message log located in the Service Desktop Manager.

4- TPS/ISE PERIPHERAL BOARD LEVEL TEST DESCRIPTIONS

The peripheral board level test descriptions are grouped by the Periph Bd Level Menu selections: PAC, SRI, and MDS .

4-1 PAC Menu Tests

The Physiological Acquisition Controller (PAC) board level diagnostics test the proper operation of the PAC. For PAC test descriptions see below:

4-1-1 Power-Up Test Description

Test time is ÷ 9 seconds. This test resets the PAC and runs the EPROM power-up tests.

- **Loopback Test** Checks communication between the PAC and SPU CPU (on IPG Board).
- **EPROM Test** Checks integrity of EPROM.
- **SRAM Test** Checks each memory location (pattern test and address test).
- **Mirror Register Test** Checks the functionality of the Mirror Register.
- **Isolated A to D Test** Checks the functionality of the A/D Converter.
- **Analog +12V Power Supply Test** Checks the +3V input voltage into the A/D Converter from the +12V supply.
- **PAL Failure Indication** Assumes that a PAL in the PAC has failed if either the Isolated A/D or the Analog +12V Power Supply Test has failed.
- **Peripheral Probe Test** Ensures that light emitting diode and Peripheral Pulse Probe is connected and working properly.

4-1-2 Resp off & Gain Test Description

Test time is ÷ 30 seconds (includes PAC reset time). This test verifies that the AGC is functioning properly. The test is composed of two sub tests: Offset Voltage Test and Voltage Gain Test.

- **Offset Voltage Test** This test verifies that the respiration voltage input to the main A/D converter can be inverted from positive to negative, or vice versa. Initially, a value of 80H is written to the Respiration circuit offset voltage so that it goes to 0V. This voltage is compared to analog ground. If the voltage is not within 5% of 0V (799H–867H), an error is sent to the Message Log.

  Next, the value of the Respiration AGC DAC is set to 4. The value of the Respiration Circuit output is compared to 0V. Any differences are calculated to make offset voltages. These offset voltages should force the Respirator voltage input to become positive or negative, depending on the offset voltage. If the Respirator voltage does not go positive if the offset is negative, or it does not go negative if the offset voltage is positive, an error is sent to the message log. If this test fails, the AGC will not proceed further.
- **Voltage Gain Test** This test verifies that the Respiration Voltage Gain is functioning properly. The test reads the current Respiration Circuit voltage. The AGC circuit is doubled, and the Respiration Circuit voltage is read once more. If the new voltage is not also doubled, an error message is sent to the message log.

4-2 SRI Menu Tests

The SRI board level diagnostics check the functionality of the Scan Room Interface (SRI) board. For individual test descriptions, see below:

4-2-1 SRI Power Up Test Description

Test time is ÷ 27 seconds. This test resets the SRI and runs the EPROM power-up tests. These tests include:

- **EPROM Test** The CPU reads contiguous16-bit word locations in the EPROM, adding one location with the next one after it. The EPROM is 64K x 16 bits, but only locations 2000H through 8000H are tested. The result is compared with the expected value.
- **SRI RAM Test** On power up, the last 512 bytes of RAM, locations FE00H–FFFFH are tested without destroying the application contents of the RAM. The SRI interrupts are disabled for this test. These locations are tested one at a time. The contents of a location are read, inverted, and written back to the location. The location is read again and compared to the value written. The original contents of the location are then written back to the location that has been tested.
- **80196 PWM Test** This test verifies that the Pulse Width Modulator is converting pulses to analog voltages correctly. (Select [Pulse Width Mod] from SRI Menu illustration for more detailed test description.)
- **Encoder Voltage Test** This test verifies that the voltage across the encoder (measured at SRI) is >4.0V. (Select **[Encoder Voltage]** from SRI Menu illustration for more detailed test description.)

4-2-2 SRI Extended RAM Test Description

Test time is ÷ 45 seconds (÷ 10 seconds with MC 68040 CPU). This test verifies the integrity of the extended RAM. The RAM on the SRI board is organized as 32K x 8 bits and resides at memory addresses 8000H through FFFFH.

The RAM is tested using a pattern test, and an address test. The values used for the pattern test are AAH, 55H, 33H, and 0FH. The address test writes a byte, whose value is associated with the address to which it is being written, to each location in RAM. This test writes the values 00H–FFH to the first 256-byte block of memory. The next 256-byte block of memory has the values 01H–FFH, with 00H in the highest location, written to it. The third 256-byte block of memory has the values 02H–FFH, with 00H and 01H in the highest locations, written to it, and so on, until all 128 256-byte blocks have been written to. This test then reads each location and checks the value read against the value of that location.

This RAM test destroys the application code of the RAM. The application code is downloaded again when you exit SRI diagnostics.

4-2-3 Encoder Voltage Test Description

Test time is ÷ 9 seconds. This test measures the voltage across the encoder with an input to the A/D converter on the 80C196. If the voltage across the encoder (measured at the SRI) is < 4.0V, an error message number is generated and sent from the SRI to the SPI.

4-2-4 Pulse Width Mod Test Description

Test time is ÷ 9 seconds. This test verifies that the Pulse Width Modulator, and the filter that converts these pulses to an analog voltage, are working. The SRI output to the cradle motor is disabled by writing a 0 (zero) to Port 0.1 of the 80196. A value is written to the PWM register in the 80196, and the PWM register is commanded to output its waveform. The voltage from the filter is then read using the A/D converter on the 80196. The values written to the PWM register, and the voltage expected from the filter output are:

PWM Register Value	PWM Duty Cycle	Expected Filter Voltage
00H	0%	0V
80H	50%	2.5V + 5%
FFH	99.6%	4.98 + 5%

4-3 MDS Menu Tests

The MDS board level diagnostics check the functionality of the Multidrop Serial Link and those peripherals on the MDS Link. The gradient board level tests are located on the Peripheral Board Level Menu. Once **[MDS]** is selected from the Submenu **[IPG]** Selection, the MDS Menu appears. For individual test descriptions, refer to the following:

4-3-1 SPI-MDS Link Description

Test time is ÷ 6 seconds. This test verifies that the SPI CPU serial port on the IPG Board is working correctly, and that the MDS Fiber Optic Link is complete.

To test the serial port, the loopback register is enabled and values (AAH, 55H, 33H, 0FH) are transmitted and read back. If these values do not match the transmitted values, the test fails.

To check the MDS Link, the loopback is disabled and four data bytes (in succession) are sent out to be received back at the SPI. These bytes have the values AAH, 55H, 33H, and 0FH. If one of these bytes does not return correctly, the test fails.

4-3-2 MDS Dev Presence Description

Test time is ÷ 7 seconds. This test verifies the operation of the SPC and SPG Boards on the MDS Fiber Optic Link. The SPI addresses only two MDS Fiber Optic Link boards at a time. If the board(s) not addressed does not transmit an error packet back to the SPI within two seconds, the test fails. An error is logged indicating which board failed (SPC or SPG).

4-3-3 RF Amp Description

Test time is ÷ 6 seconds. This test verifies the operation of the MDS Link and the presence of the ERBTEC RF Amplifier. To verify its presence, the Command Register is addressed and read by the SPI via the MDS. No additional tests are used.

4-3-4 TR Dyn Bd Reg Description

Test time is ÷ 9 seconds. This test verifies that the TR/Dynamic Disable function is in working order. This test is comprised of the following subtests:

- **Presence Test** The SPI on the IPG Board addresses the command register on the TR/Dynamic Disable Board. If the board is not present, an error is logged.
- **Coil Configuration Register Test** The coil configuration register is tested by writing/readback of patterns AAH, 55H, 33H, and 0FH.
- **Fault Reporting Disable Register Test** The Fault Reporting Disable Register is tested by writing/readback of patterns AAH, 55H, 33H, and 0FH.
- **TR Command Register Test** The TR Command Register is tested by writing/readback of bits 00 (Enable), 01 (Service), 02 (Reset), and 07 (Reset Status Registers). After each bit is read, the bit is checked to see if it has been reset.

- **Register Reset Tests** The Dynamic Disable and TR Driver Board has two status registers: the TR/ID Status Register and the Dynamic Disable Status Register. These status registers are reset to 00H by setting bit 07 of the TR Command Register. After bit 07 of the TR Command Register is set, the Status Registers are checked to see if they are reset. Also, the Coil Configuration and Fault Reporting Disable Registers are each written with pattern FFH. Then, bit 02 of the TR Command Register is set, and these registers are checked to see if they have gone to 00H.

4-3-5 TR Dyn Pwr Supl Description

Test time is ÷ 5 seconds. These tests verify the operation of the three power supplies (+15V, –15V, and +1000V) associated with the Dynamic Disable and TR Driver Board. The status of these power supplies is indicated by the fault register bits 05 (+15V), 06 (–15V), and 07 (+1000V). These bits are each sampled five times (with approximately 65 milliseconds between each sample). A failure is displayed if one of the power supplies is indicated as having failed five times in a row.

4-3-6 Power Monitor Description

Test time is ÷ 6 seconds. This test verifies the operation of the MDS Fiber Optic Link, the operation of the Power Monitor and the presence of the Power Monitor. To verify its presence, the SPI on the IPG Board will request the status of the Power Monitor from the CMB. If no status is returned, the SPI assumes that it is not connected to the MDS. The test below is used to further test the Power Monitor.

- **Data Test** The MDS diagnostics verify that the power monitor can accept data; therefore, it sends the maximum values for the amplitude, pulse width, and duty cycle of the signal that is picked up by the head coil, the body coil, and the surface coil, respectively. These data are followed by a checksum to verify data integrity.

4-3-7 Multicoil Bd Reg Description

Note

This test is present only on systems with the multicoil option.

Test time is ÷ 5 seconds. This test verifies the operation of the MDS Fiber Optic Link and the presence and operation of the Multicoil Board. To verify its presence, the Command Register of the Dynamic Disable/TR Driver Board is read. To verify the functionality of the registers, a set of Multicoil Board Register tests are run simultaneously with patterns AA, 55, 33, OF. Any errors are logged to indicate the failing register.

4-3-8 Digital Tuning Board Exponential Monitoring Test Description

Test time is ÷ 5 seconds. This test is the primary means of functionally testing the GRAM Digital Tuning Board. The test uncovers other problems in the Gradient Driver Subsystem, but focuses mainly on the Digital Tuning Board.

This test is useful only on Signa Horizon HiSpeed and Signa Horizon EchoSpeed systems. A GRAM must be present for Digital Tuning Boards to be present. In addition, each AIS must have MIF, GRAM, and GASM. ASM boards are not required.

For more information, see the nonproprietary procedure for Digital Tuning Board Exponential Monitoring

This button is present only on Signa Horizon systems with a GRAM and Digital Tuning Boards.

4-3-9 Spectro RF Amp Description

Note

This test is present only on systems with the spectroscopy option.

Test time is ÷ 6 seconds. This test verifies the operation of the MDS Fiber Optic Link and the presence and operation of the Spectro RF Amplifier Interface Board. To verify its presence, the Command Register is addressed and read by the SPI on the IPG Board via the MDS. The test below is used to verify the functionality of the Spectro RF Amplifier.

- **Spectro RF Amplifier Interface Board Register Test** The register is read for a specific pattern. The pattern indicates that the RF amplifier is connected through P2201 to the Dynamic Disable Module.

  Note

  The GAP Framing Error / Clock Stop test logs the following error whenever it is run (except the first time after a TPS/ISE chassis reset, and this is the FIRST test to run). For example:

  ```
  2222304

  GAP Framing Error/Clock Stop Test Failure.

  The GAP failed to detect and/or report a Framing error
  during each pass of this test.

  Framing Errors Detected: 0
  Framing Errors Expected: 1
  ```

This test is part of the MDS Peripheral diags set.

4-4 Gradient Driver Tests

Since Gradient Driver Tests have a greatly expanded capability, a separate document has been created for them. For more information regarding these new hybrid diagnostics, see the procedure for Gradient Driver Tests.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 29, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	January 5, 1999	K. Keshena	Updated styles.
2	February 22, 1999	F. Fiore	Edits based on bay validation.